higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

LOGIC SYSTEMS N5

8 APRIL 2019

This marking guideline consists of 7 pages.

Copyright reserved

QUESTION 1

A	B	C	J_A	K_A	J_B	K_B	J_C	K_C
1	1	0	Ø	0	Ø	1	1	Ø
1	0	1	Ø	0	0	Ø	Ø	1
1	0	0	Ø	1	1	Ø	1	Ø
0	1	1	0	Ø	Ø	0	Ø	1
0	1	0	0	Ø	Ø	1	1	Ø
0	0	1	1	Ø	1	Ø	Ø	1
1	1	1	Ø	0	Ø	0	Ø	1
0	0	0	1	Ø	1	Ø	0	Ø

$Q_n \rightarrow Q_n + 1$	J	K
0 → 0	0	Ø
0 → 1	1	Ø
1 → 0	Ø	1
1 → 1	Ø	0

	00	01	11	10
0	1	0	Ø	Ø
1	1	0	Ø	Ø

$J_A = /B$

	00	01	11	10
0	Ø	Ø	0	1
1	Ø	Ø	0	0

$K_A = /B/C$

	00	01	11	10
0	1	Ø	Ø	1
1	1	Ø	Ø	0

$J_B = /A + /C$

	00	01	11	10
0	Ø	1	1	Ø
1	Ø	0	0	Ø

$K_B = /C$

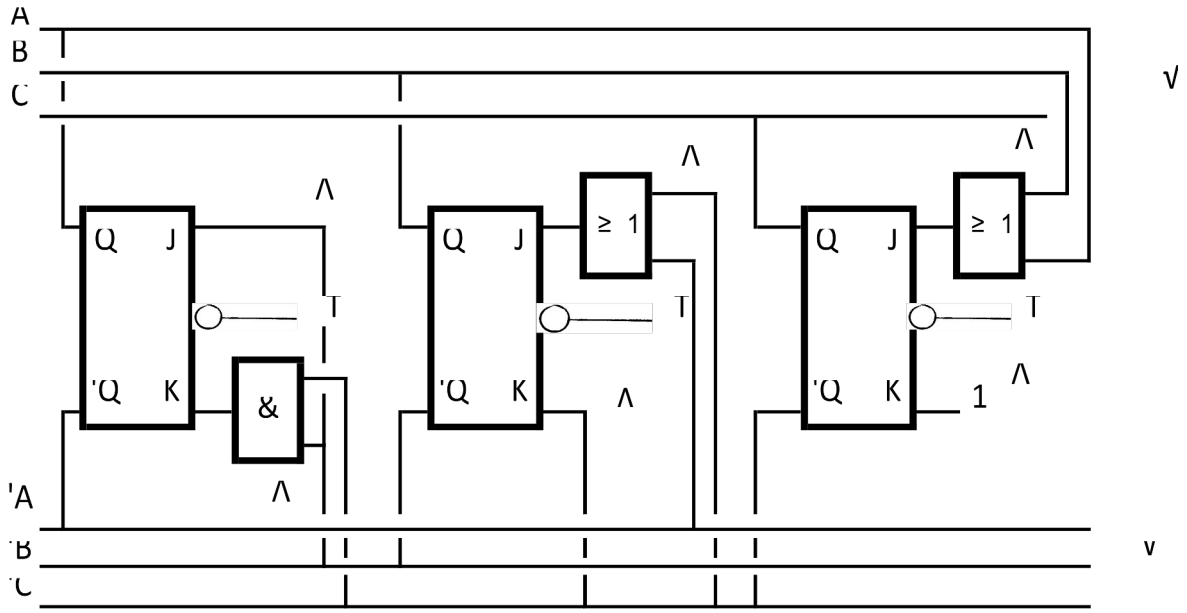
	00	01	11	10
0	0	1	1	1
1	Ø	Ø	Ø	Ø

$J_C = A + B$

	00	01	11	10
0	Ø	Ø	Ø	Ø
1	1	1	1	1

$K_C = 1$

(7)

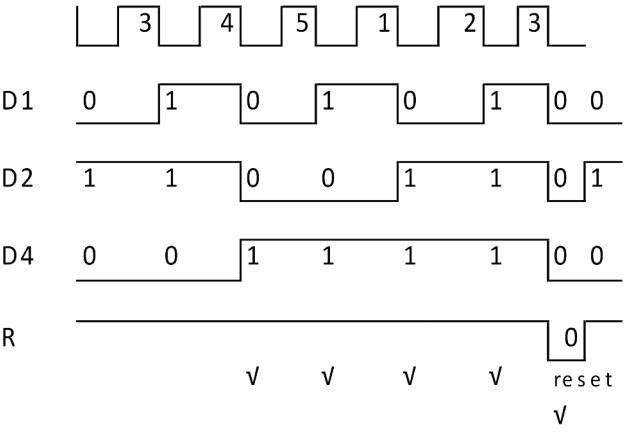


(8)

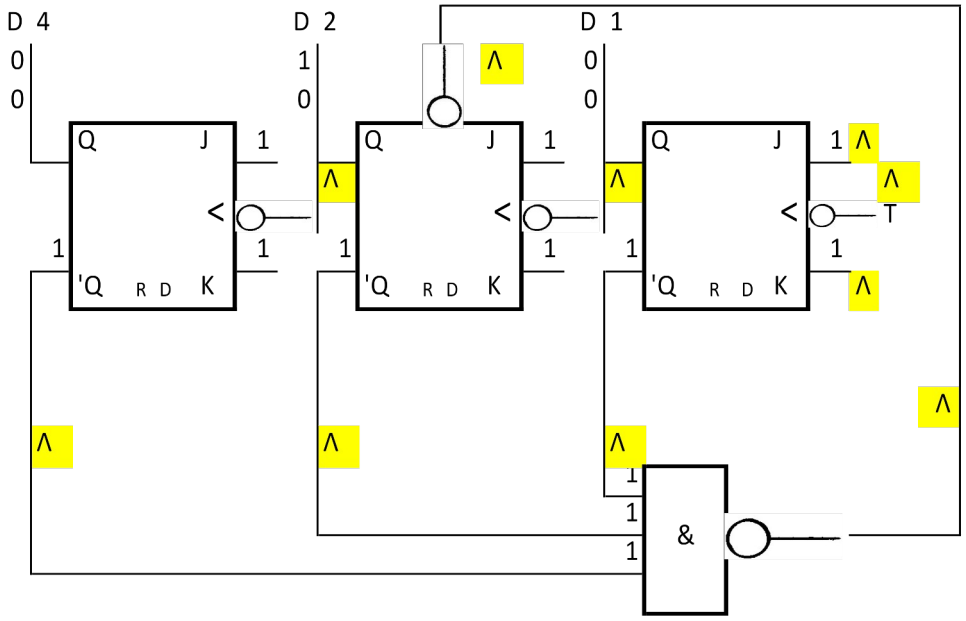
[15]

QUESTION 2

	D4	D2	D1	T	T
	0	1	0	0	
	0	1	1	1	√
	1	0	0	2	√
	1	0	1	3	√
	1	1	0	4	√
	1	1	1	5	√
Reset	0	0	0	6	
	0	1	0		



(10)

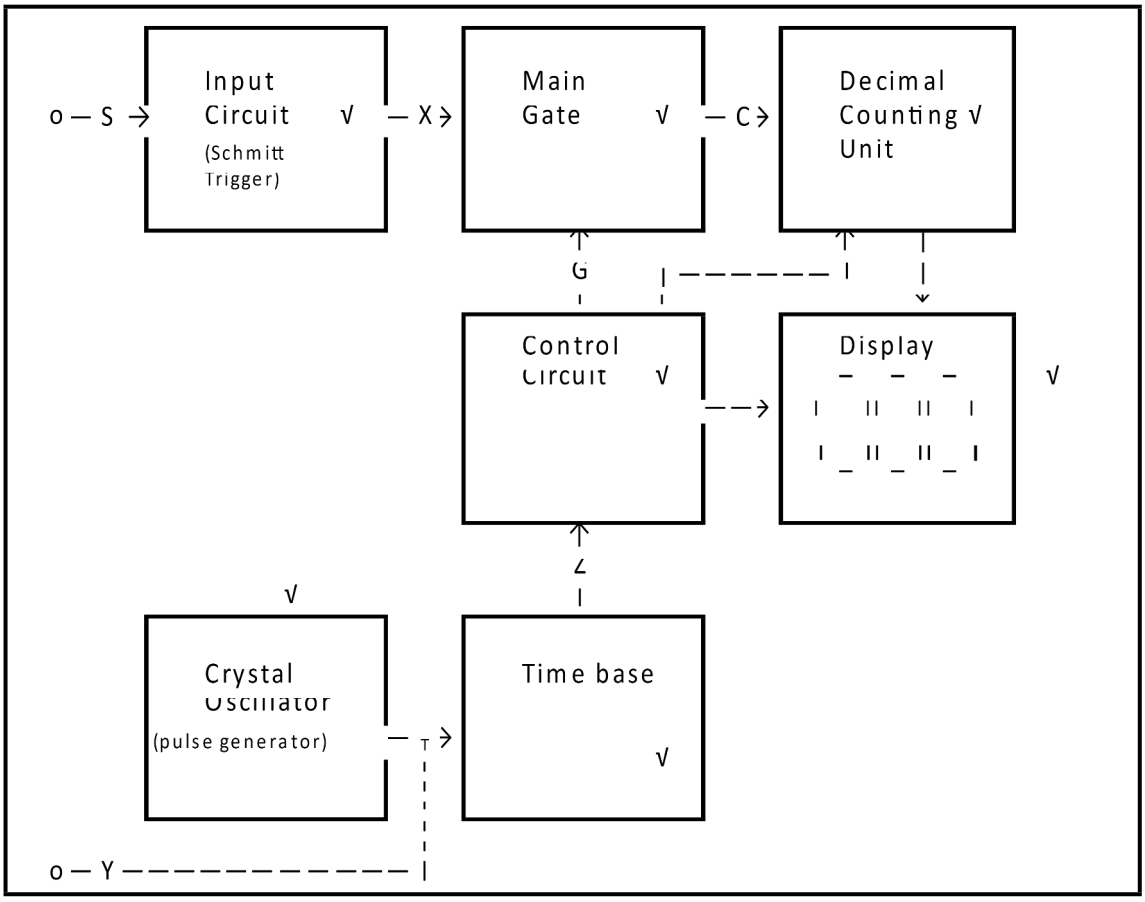


Λ = 1/2 mark 10 * 1/2 = 5 marks

(5)
[15]

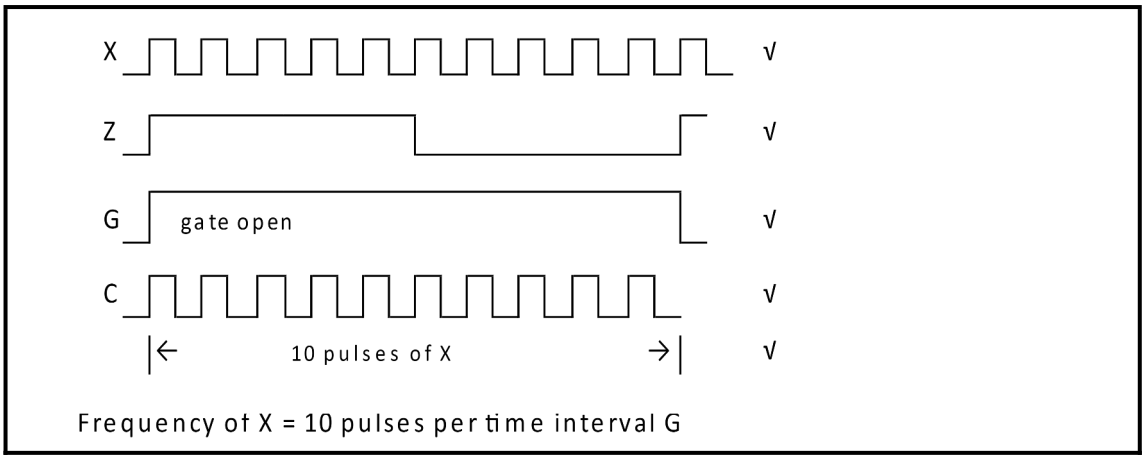
QUESTION 3

3.1



(7)

3.2



(5)
[12]

QUESTION 4

4.1 False
4.2 True
4.3 True
4.4 True
4.5 False
4.6 True
4.7 True
4.8 False
4.9 True
4.10 False
4.11 False
4.12 True
4.13 False
4.14 False
4.15 True

(15 × 1) **[15]**

QUESTION 5

5.1 K
5.2 M
5.3 O
5.4 N
5.5 I
5.6 J
5.7 L
5.8 F
5.9 C
5.10 G
5.11 A
5.12 H
5.13 E
5.14 B
5.15 D

(15 × 1) **[15]**

Copyright reserved

QUESTION 6

6.1 SRAM memory is a type of semiconductor memory where the word static indicates that it does not need to be periodically refreshed while power is maintained. (2)

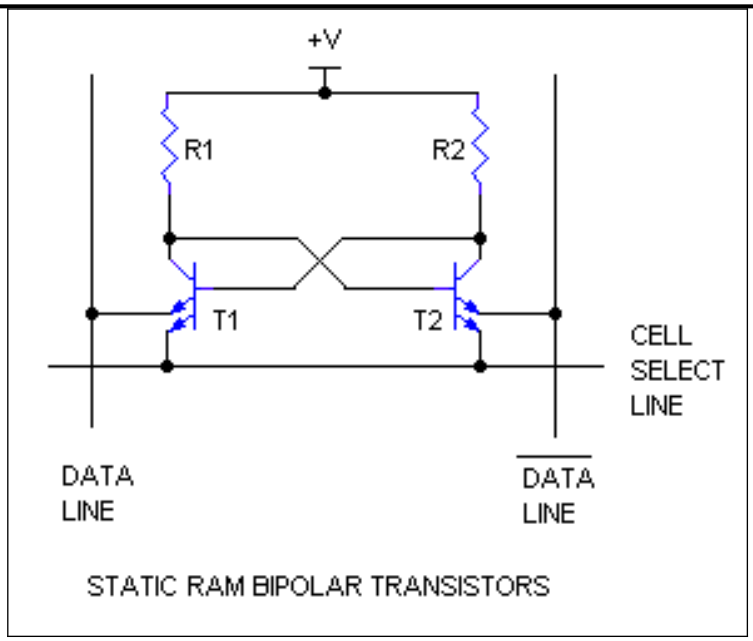
6.2 Volatile memory is computer memory that requires power to maintain the stored information, unlike non-volatile memory which does not require a maintained power supply. (2)

6.3 DRAM memory is a type of random access memory that stores each bit of data in a separate capacitor within an integrated circuit. (2)

6.4 Non-volatile memory (NVM) is computer memory that can retain the stored information even when not powered. (2)

Examples are ROM, flash memory, hard disks, floppy disks, magnetic tape, optical discs and early computer storage methods such as paper tape and punch cards. (Any 2 × 1) (2)

6.5



(1 mark each for any 4 correct labels) (4)

[14]

QUESTION 7

7.1 Character
7.2 Spins
7.3 Hammer
7.4 Presses
7.5 Character
7.6 Print
7.7 Striking
7.8 Paper
7.9 Matrix
7.10 Rod
7.11 Electromagnet
7.12 Chain
7.13 Character
7.14 Hammer

(14 × 1) **[14]**

TOTAL: 100

Copyright reserved